INCH-POUND
MIL-G-24745(SH)
4 September 1990

MILITARY SPECIFICATION

GOVERNORS, STEAM DRIVEN PUMP, PRESSURE REGULATING, NAVAL SHIPBOARD, 600 POUNDS PER SQUARE INCH STEAM SERVICE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements applicable to the design and construction of self-contained piston operated pump pressure regulating governors.

1.2 Classification. Governors shall be of the following types and sizes as specified (see 6.2):

1.2.1 Types.

Type CP - Constant pressure.
Type CPASP - Constant pressure with automatic stop.

1.2.2 Sizes. The main steam governing valve size shall be compatible with the auxiliary turbine it serves.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4820
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:39Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500).
QQ-S-763 - Steel Bars, Wire, Shapes, and Forgings, Corrosion Resisting.

MILITARY

MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.
MIL-S-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-R-17131 - Rods and Powders, Welding, Surfacing.
MIL-G-21032 - Gaskets, Metallic-Asbestos, Spiral Wound.
MIL-P-24503 - Packing Material, Graphitic, Corrugated Ribbon or Textured Tape and Preformed Ring.
MIL-C-24707 - Castings, Ferrous, General Specification for.
MIL-C-24707/2 - Castings for Pressure Containing Parts Suitable for High Temperature Service.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-248 - Welding and Brazing Procedure and Performance Qualification.
MIL-STD-271 - Requirements for Nondestructive Testing Methods.
MIL-STD-278 - Welding and Casting Standard.
MIL-STD-798 - Nondestructive Testing, Welding, Quality Control, Material Control and Identification and HI-Shock Test Requirements for Piping System Components for Naval Shipboard Use.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 202640-20T01:39Z
Check the source to verify that this is the current version before use.

AMERICAN NATIONAL STANDARDS INSTITUTE
B16.5 - Pipe Flanges and Flanged Fittings.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS
A 193 - Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High Temperature Service. (DoD adopted)
A 194 - Standard Specification for Carbon and Alloy Steel Nuts for Bolts for High-Pressure and High-Temperature Service. (DoD adopted)
A 216 - Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High Temperature Service. (DoD adopted)
B 103 - Standard Specification for Phosphor Bronze Plate, Sheet, Strip, and Rolled Bar. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.2 General requirements.

3.2.1 Reliability. The principle of reliability is paramount and no compromise of this principle shall be made with any other basic requirements of design. This specification is intended to obtain governors of such design that they will operate over a long period of years with a minimum of servicing. Where wear or erosion is unavoidable the parts subjected to such wear or erosion shall be of the best materials available for the purpose in order to reduce these detrimental effects to a minimum. The design and construction of all governors shall be the most compact consistent with the following requirements:

(a) Reliability.
(b) Accessibility for repair.
(c) Resistance to wear or corrosion.
(d) Economy.
(e) Satisfactory operation when inclined as follows:

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:39Z
Check the source to verify that this is the current version before use.

(1) Up to 15 degrees to either side (permanently inclined).
(2) With the ship rolling up to 45 degrees from the vertical to either side.
(3) With the ship pitching 10 degrees up and down from the normal horizontal plane.

3.2.2 Shock. The governors shall withstand the high-impact shock test specified in MIL-S-901 (see 4.6.2).

3.2.3 Capacity rating. Each governor body and device shall be designed to suit the highest working pressure and steam capacity (see 6.2), on which it is to be used, and shall be sufficiently rugged to withstand stresses which may be transmitted via the piping installations arising from vibration, expansion and contraction.

3.2.4 Suitability. Governors shall be suitable for use with centrifugal or rotary pumps.

3.2.5 Access for repair. The construction of all governor valves and devices shall permit easy access to all parts for overhaul and repair. Wearing parts, with the exception of integral seats, shall be removable without removing the governor from the line.

3.2.6 Adjustment. Each governor shall be suitable for the particular pump discharge pressure and range of adjustment as specified (see 6.2).

3.2.7 Steam governing valves.

3.2.7.1 Pressure drop. The maximum pressure drop through the governor valve at the required flow capacity shall not exceed 30 lb/in^2.

3.2.7.2 Manual control. Each steam governing valve shall be fitted with a simple, manually operated feature by which the automatic governing action may be rendered inoperative and allow full flow of steam through the valve. This feature shall consist of a handwheel and stem acting directly on the main steam valve. An indicating device marked "Auto" and "Open" shall be fitted.

3.2.7.2.1 Packing box assembly. A packing box assembly shall be provided to seal against leakage along the manual operator stem. Sealing bellows shall not be used.

3.2.7.2.2 Packing. Packing shall be designed for sustained operation at the temperature and pressure conditions to which it will be subjected.

3.2.7.3 Contamination protection. All governors shall be compatible with the actuating fluid and compatible with the required pressure and temperature (see 6.2). When the actuating medium is any fluid other than air or fresh water the valves shall be so designed that rupture of any diaphragm, or leakage through any joint cannot result in contamination of the steam driving the pump unit.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:39Z
Check the source to verify that this is the current version before use.

3.2.7.4 Bodies. Governor valves shall have straight-through style main bodies (flanges on common centerline) unless otherwise specifically approved by NAVSEA or the agency concerned.

3.2.7.5 Connections. The steam valves shall have flanged connections throughout, including spring chambers, unless otherwise specified. Inlet and outlet flanges shall form part of the valve body. All flanges shall be drilled in accordance with ANSI B16.5 (see 6.2).

3.2.7.5.1 Gasket surfaces. All gasket surfaces shall be furnished with a serrated concentric or phonographic finish with roughness not exceeding 500 roughness height rating (RHR) and a minimum of 40 cuts per inch of face width.

3.2.7.5.2 Flange bolting. Flange bolting shall straddle the vertical centerline. Where an odd number of bolts is used, the vertical centerline at the top of the flange shall be straddled.

3.3 Description of types.

3.3.1 Type CP - constant pressure. This governor shall be designed to maintain a constant predetermined pressure at a selected point in the pump discharge line by control of the steam supply to the pump driver, regardless of variation in steam pressure and pump load.

3.3.2 Type CPASP - constant pressure with automatic stop. The type CPASP governor is the basic type CP with additional automatic stopping features. This governor shall be designed and arranged to positively and automatically stop a pump when the pressure in the actuating line to the stopping device falls below a certain low predetermined pressure, and shall not automatically restart the pump even though the actuating line pressure is reestablished (see 6.2). This governor shall not incorporate a gauging device.

3.3.2.1 Stopping pressure. The stopping device shall be adjustable as to stopping pressure.

3.4 Design.

3.4.1 Design features. The basic design features specified herein are desirable; however, consideration will be given to governors differing in design from the features described (see 6.2) provided it is demonstrated to NAVSEA or the agency concerned that none of the general requirements and types enumerated in 3.2 and 3.3 are sacrificed and the unit meets all of the first article requirements in 4.3.

3.4.2 Basic design. This basic design shall include a piston-operated main steam governing valve controlled by an auxiliary valve, the auxiliary valve being operated by an adjustable external spring acting against a diaphragm and balanced by the pressure in the actuating line (see 6.3 and appendix A).

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:39Z
Check the source to verify that this is the current version before use.

3.4.3 Auxiliary valve. The auxiliary valve shall be single-seated spring loaded type with removable valve seats. The auxiliary valve chamber shall be self-draining to prevent accumulation of water around the auxiliary valve spring.

3.4.4 Main valve. The main steam valve for CPASP type governors shall be single seated spring loaded type. The valve seat shall be integral with the valve body and shall be faced with cobalt chromium composition facing. The main steam valve for CP type governors shall meet the above requirements but may be of a double seat design.

3.4.5 Access. The piston operated design shall have cylinder liners which are readily removable. The main valve shall not be integral with its operating piston.

3.4.6 Automatic moving parts. No packing glands, stuffing boxes, metal bellows or syphons shall be used on any automatically moving parts.

3.5 Materials.

3.5.1 Materials. Materials for steam governing valves shall be as specified in table I. Material for parts not shown in table I shall be of the quality best suited for the purpose intended, and shall be subject to review and approval by NAVSEA or the agency concerned. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product.

TABLE I. Materials. 2/

Parts	Specification	Material
Body, bonnet, and bottom cap	ASTM A 216, grade WCB	Carbon steel for temp. up to 775°F
	MIL-C-24707 and MIL-C-24707/2, grade WC9	Chrome-moly alloy for temp. up to 850°F
Diaphragms in contact with steam	Inconel X-750	Nickel-chrome alloy
Diaphragms in contact with oil or fresh water	Inconel X-750	Nickel-chrome alloy
	ASTM B 103	Phosphor bronze
	QQ-N-286	Nickel-copper aluminum alloy

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:39Z
Check the source to verify that this is the current version before use.

TABLE I. Materials 2/ - Continued.

Parts	Specification	Material
Diaphragms in contact with sea water	QQ-N-286	Nickel-copper aluminum alloy
Main valve plug and seat 1/	AISI 410 or 420 stainless steel	Chrome-iron alloy hardened
Auxiliary valve plug and seat 1/	AISI 420 or 440C stainless steel	Chrome-iron alloy hardened
Main and auxiliary valve stem	QQ-S-763	Corrosion-resistant steel
Cylinder liners and guide bushings	AISI 410, 420 or 440C	Chrome-iron alloy hardened
Pressure containing fasteners	ASTM A 193	Steel grade B7
	ASTM A 194	Steel grade 2H
Springs main and pilot valve	Inconel X-750	Nickel-chrome alloy
Top and bottom cap gaskets	MIL-G-21032 with filler in accordance with MIL-P-24503	Spiral wound type

1/ Main valve trim materials. Main valve trim (defined as consisting of the seat or seat ring, and plug and the guide posts and bushings) materials shall be selected from the following:

(a) Stellite - shall consist of either wrought Stellite 6B, cast Stellite 6, or an inlay of Stellite (not less than 3/32 inch thick for main seat and disc surfaces). Where inlays are used, welding rods shall be in accordance with type MIL-RCoCr-A of MIL-R-17131.

(b) Hardened corrosion-resistant steel - Hardened corrosion-resistant steel plug (400 series or 17-4 PH) and Stellite seat or seat ring. Guiding surfaces shall be hardened corrosion-resistant steel or Stellite.

Non-galling grades of materials shall be chosen to prevent galling between rubbing surfaces. A difference in hardness of at least 100 points Brinell shall be maintained between the rubbing surfaces of the guide bushings and posts. This requirement does not apply if either the guide bushings and posts are Stellited or if the hardness of either exceeds 450 Brinell.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:39Z
Check the source to verify that this is the current version before use.

2/ Compliance with the material requirements of this specification does not warrant the acceptance of a design. The design features (i.e., mechanical properties) shall be suitable for extended shipboard use as determined by the Navy.

3.5.2 Material variation. The use of materials differing in properties from those specified for a given application will be considered when the contractor shows the necessity for such substitution, the material proposed is readily obtainable, and provided that the material is satisfactory to NAVSEA or the agency concerned.

3.5.3 Welding. Welding shall be in accordance with MIL-STD-248 and MIL-STD-278.

3.5.4 Threaded parts. Threaded parts shall conform to FED-STD-H28. A suitable anti-seize compound shall be used on threaded parts subjected to a temperature above 300 degrees Fahrenheit (°F).

3.5.5 Prohibited fasteners. The use of tap bolts or cap screws for bottom cap or bonnet joints is prohibited unless specifically approved by NAVSEA or the agency concerned.

3.5.6 Nondestructive test. All pressure castings and weldments shall meet the nondestructive test requirements of MIL-STD-278 and MIL-STD-271.

3.6 Performance requirements.

3.6.1 Performance. The performance requirements described below are based upon the overall characteristics of the governor/turbine/pump combination. The governor rise characteristic required to provide proper operation of the complete system shall be determined by tests on each new pump/driver combination (see 6.2). The governor manufacturer shall then test all subsequent governors to assure repeatable rise characteristics. These subsequent tests may be run without the driver/pump.

3.6.1.1 Governors for boiler feed pump (BF) duty.

3.6.1.1.1 Discharge pressures at shut-off. When centrifugal pumps are operating under normal steady steam pressure controlled by BF duty governors, the pump discharge pressures at shut-off shall be not less than 2.5 percent nor more than 4.0 percent above the rated discharge pressure of the pumps at rated capacity. This rise in pressure shall be continuous; that is, at no point of the curve from rated capacity to shut-off shall there be a drop in the characteristic. (This requirement is based on pumps having constantly rising head-capacity curves with shut-off, at constant rated speed, not less than 15 percent greater than rated discharge pressure.)

3.6.1.1.2 Pressure capacity curves. The pressure capacity characteristic curves of a pump with a governor in control, when determined from rated capacity to shut-off and from shut-off to rated capacity, shall not vary at any given

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:39Z
Check the source to verify that this is the current version before use.

capacity by an amount in excess of 0.67 percent of the rated discharge pressure for pumps rated at 750 lb/in^2 or less, and 1.0 percent for pumps rated at more than 750 lb/in^2.

3.6.1.1.3 Pressure fluctuations. When a pump is operated under steady conditions, the constant pump discharge pressure for which a governor is set shall not vary more than plus or minus 0.67 percent of the governor setting. The maximum momentary variation in pump discharge pressure due to sudden major changes in capacity shall not exceed plus or minus 10 percent of the rated pump discharge pressure, and such fluctuations shall settle down to normal within 10 seconds.

3.6.1.1.4 Discharge pressure variations. During a gradual reduction in steam pressure of 20 percent, or gradual increase of 20 percent, the pump discharge pressure shall not vary more than plus or minus 1.5 percent, with turbine overload nozzle open.

3.6.1.2 Governors for fuel oil pump (FO) duty.

3.6.1.2.1 Discharge pressure variations. When pumps are operating under steady conditions controlled by FO duty governors, the pump discharge pressures shall not vary more than plus or minus 2 percent.

3.6.1.2.2 Governor stability. Under steady normal operating conditions with any setting of pump discharge pressure, the pump discharge pressure shall rise not less than 2.8 percent nor more than 5.7 percent of the governor setting during gradual changes from full load to 25 percent load and vice versa, and under those conditions the governor shall be stable from full load to minimum load and vice versa.

3.6.1.2.3 Discharge stability. During gradual reduction in steam pressure of 20 percent and vice versa, the pump discharge shall not vary by more than 2.8 percent of the rated discharge pressure.

3.6.1.2.4 Momentary variation. Maximum momentary variation in pump discharge pressure due to sudden major changes in capacity with steady steaming conditions shall not exceed plus or minus 10 percent of the governor pressure setting, and such fluctuations shall settle down to normal within 10 seconds.

3.6.1.3 Governors for lube oil pump (LO) and fire pump (FS) duty.

3.6.1.3.1 Pump discharge pressure variations. Under steady operating conditions the pump discharge pressure shall not vary more than 1 lb/in^2, total for LO service, and plus or minus 2 percent, or 5 lb/in^2, whichever is greater, for FS service.

3.6.1.3.2 Pump discharge tolerances. Under normal operating steam conditions, with any setting of the governor, the pump discharge pressure shall be maintained within a tolerance of plus or minus 5 lb/in^2 during gradual changes from 25 percent load to full load and vice versa; also during a gradual reduction in steam pressure of 20 percent and vice versa the same pump discharge pressure tolerances shall be maintained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:39Z
Check the source to verify that this is the current version before use.

3.6.1.3.3 Maximum momentary variation. Maximum momentary variation due to sudden major changes in capacity with steady steam conditions shall not exceed plus or minus 10 percent of the pump discharge pressure for pressures of 100 lb/in^2 or above, or plus or minus 10 lb/in^2 for lower pressures; and such fluctuations shall settle down to normal within 10 seconds.

3.7 Identification plates.

3.7.1 Each steam governing valve body shall have the following information stamped or cast thereon, or on a metal identification plate permanently attached thereto:

(a) The governor type and normal size.
(b) The rated steam pressure and temperature (see 6.2).
(c) The manufacturer's type or model designation and serial number.
(d) The trademark or name of the manufacturer.
(e) An arrow indicating the direction of steam flow.
(f) The standard Navy stock number, if assigned.
(g) Contract or order number.
(h) Pump discharge pressure range.

3.7.2 Identification plates shall be type A or B in accordance with MIL-P-15024, except that plates of plastic or aluminum are not acceptable. Plates shall be secured to the equipment with corrosion resistant metal screws, drive screws, or rivets.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program (see 6.3). The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:39Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection is required for each type of governor. First article inspection shall consist of the examination of 4.5 and the tests of 4.6.

4.4 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.5 and the tests of 4.6.1 and 4.6.4.

4.5 Examination. The valve shall be disassembled. A visual and dimensional examination shall be performed to determine conformance with the requirements of this specification and drawings.

4.6 Tests. Test procedures shall consist of the tests specified in 4.6.1 through 4.6.5.

4.6.1 Nondestructive tests. Nondestructive tests shall be as specified in MIL-STD-278 and in accordance with MIL-STD-271. Acceptance criteria shall be in accordance with MIL-STD-278. This shall include RT, MT/PT, pressure, and visual testing.

4.6.2 High-impact shock test. The valve shall be subjected to the high impact mechanical requirements for grade A, class I of MIL-S-901. MIL-STD-798 shall be used as guidance for test procedures.

4.6.3 Vibration test. The valve shall be vibration tested in accordance with type I of MIL-STD-167-1.

4.6.4 Performance test. The valve shall be subjected to the tests specified in 4.6.4.1 through 4.6.4.4. The delivered pressure shall at all times hold within the limits of plus or minus 5 percent or plus or minus 2 lb/in^2, whichever is greater. There shall be no evidence of hunting, chattering or any other unstable or unsatisfactory operation of the valve during any portion of these tests. The test specified in 4.6.4.1 and 4.6.4.2 shall be conducted with the valve set at the upper and lower limits of the set pressure range. Tests shall be done on a suitable pump with turbine.

4.6.4.1 Flow-drop test. The flow shall be varied from lock-up to the maximum flow rating of the valve in not more than 15 seconds and from the maximum flow rating to lock-up in not more than 15 seconds. This test shall be conducted under the following sets of conditions:

(a) Maximum inlet pressure - lowest set pressure.
(b) Minimum inlet pressure - lowest set pressure.
(c) Minimum inlet pressure - highest set pressure.
(d) Maximum inlet pressure - highest set pressure.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:39Z
Check the source to verify that this is the currentversion before use.

The maximum inlet pressure for these tests shall be equal to the rating of the valve and the minimum inlet pressure for these tests shall be equal to 75 percent of the rating of the valve.

4.6.4.2 Inlet pressure transient. With a constant flow demand equal to 10, 50, and 100 percent of the maximum, the inlet pressure shall be varied from 75 percent to 100 percent of the inlet rating in not more than 5 seconds and from 100 percent to 75 percent of the inlet rating in not more than 5 seconds.

4.6.4.3 Endurance test. Following the tests of 4.6.4.1 and 4.6.4.2, the valve shall be subjected to a 20 hour operational test to check functioning and performance.

4.6.4.4 Each governor valve body assembly shall be hydrostatically tested in accordance with ANSI B16.5.

4.6.5 Post test examination. After the completion of the tests specified in 4.3, the test valve shall be disassembled and visually examined. Any damage, excessive wear, or signs of galling or pitting shall be cause for rejection.

4.7 Inspection of packaging. Sample packages and packs and the inspection of the preservation, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.6.)

5.1 Packaging. The packaging (preservation, packing and marking) requirements for governor valves and manuals (see 6.4), shall be in accordance with MIL-V-3 for the level (A, C or commercial) of preservation and level (A, B, C, or commercial) of packing, marking, including packaging acquisition options therein as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Governors covered by this specification are intended for use aboard naval ships for steam driven pumps, steam service 600 max and 850°F maximum.

6.2 Acquisition requirements. Acquisition documents must specify the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:39Z
Check the source to verify that this is the current version before use.

(a) Title, number, and date of this specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) Type and size required (see 1.2).
(d) If first article test is required (see 3.1).
(e) Required pump discharge system pressure and range of adjustment (see 3.2.6).
(f) Type and pressure and temperature limits of the governor actuating fluid (see 3.2.7.3).
(g) Whether steam valves are to have other than flanged connections (see 3.2.7.5).
(h) Automatic stopping pressure, for a type CPASP governor (see 3.3.2).
(i) Valve capacity in pounds of steam per hour (see 3.2.3).
(j) Pump service (that is, main feed, fuel oil service, etc.) (see 3.6.1).
(k) Steam inlet pressure and temperature (see 3.7.1).
(l) Level of preservation and packing, and packaging options required (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.4.2 and appendix A	DI-DRPR-80651	Engineering drawings	-----
4.1.1	DI-R-4803	Inspection system program plan	-----

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract. Technical content should include the requirements of appendix B.

Source: https://assist.dla.mil -- Downloaded: 202640-20T01:39Z
Check the source to verify that this is the currentversion before use.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Repair parts and tools.

6.7.1 On board. Each set of on board repair parts should include the following as applicable, plus any additional parts considered by the manufacturer to be essential in successful servicing of the governors:

(a) 100 percent complete replacement for all governors per ship:

(1) Diaphragms.
(2) Springs exposed to steam.
(3) Special packing, seals.
(4) Gaskets.
(5) Auxiliary valves.
(6) Auxiliary valve seats.
(7) Piston rings.

(b) One part for each four governors or fraction thereof, per ship:

(1) Springs of each size and type not exposed to steam.
(2) Cylinder liners of each size and type.

6.7.2 Tools. A complete set of special tools required for the maintenance and repair of the governors should be included in each set of on board repair parts. No ordinary commercial tools or wrenches shall be included. Each tool or wrench must be indelibly marked for the purpose intended. Special tools are defined as those tools not listed in the Federal Supply Catalog. (Copies of this catalog may be consulted in the office of the Defense Contract Administration Services Management Area (DCASMA)).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:39Z
Check the source to verify that this is the current version before use.

6.7.2.1 Special tools and wrenches should be listed in the technical manual and included in the lists of material on sectional assembly drawings.

6.7.3 Stock. The selection, stock numbering, and quantities of repair parts for stock should be determined and processed in accordance with MIL-P-15137.

6.8 Subject term (key word) listing.

Diaphragms
Discharge pressure
Flanges
Flange bolting
Gaskets
Pressure capacity
Valves

Preparing activity:
Navy - SH
(Project 4820-N036)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:39Z
Check the source to verify that this is the current version before use.

APPENDIX A

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical content requirements that shall be included on drawings when required by the contract or order. This appendix is mandatory only when data item description DI-DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DRAWING CONTENT

30.1 Drawing content. In addition to the drawing content required by the data ordering document, the following unique features shall be included:

(a) Accurately scaled sectional assembly which clearly depicts the design and construction of the valve.
(b) Bill of material listing specification, grade condition, and any other data required to fully identify the properties of the materials proposed.
(c) Details of the seat, disc, and stem assembly and all other replaceable internal trim.
(d) Drawings of the pressure-containing envelope (body, bonnet, and bottom cap) giving dimensions which control compression of the spiral-wound gaskets. This is to assure that where rematching is necessary to repair the gasket sealing surfaces on these parts, that compensating cuts can be accurately made to restore original gasket compression. This drawing shall also specify the dimensional limits of such corrective remachining within which function of the valve remains unaffected.
(e) Recommended assembly torques, or equivalent procedures, for making up all joints and threaded assemblies.
(f) Tabulation of required gasket characteristics including all dimensions (with tolerances) and load versus compression characteristics (with tolerances).
(g) Marked areas to be radiographic, magnetic particle, or dye penetrant inspected.
(h) Detail drawing for all special tools.

30.2 Certification data sheets. When required by the contract or order, the certification data sheets shall include the following:

(a) Application description including:

(1) Steam inlet pressure and temperature.

Source: https://assist.dla.mil -- Downloaded: 202640-20T01:39Z
Check the source to verify that this is the currenversion before use.

(2) Pump service (that is, main feed, fuel oil service, etc).
(3) Valve capacity in pounds of steam per hour.
(4) Automatic stopping pressure, for a type CPASP governor.
(5) Required pump discharge system pressure and range of adjustment.
(6) Valve description.
(7) Required maximum capacity under specified conditions.
(8) Rated maximum capacity under specified conditions.
(9) Fail-open capacity (for purposes of relief valve sizing).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:39Z
Check the source to verify that this is the current version before use.

APPENDIX B

TECHNICAL MANUAL TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical content requirements that shall be included in technical manuals when required by the contract or order. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. TECHNICAL MANUALS

30.1 Content. In addition to the general requirements for technical manuals, the following shall be included as part of the contents:

(a) Drawings for the valve (including certification data sheet). Drawings shall be supplemented by additional illustrations where necessary to adequately illustrate operation and maintenance. These additional illustrations may consist of blowouts, partial or full sections, special tools, etc., and may eliminate extraneous lines and details to clarify the interaction of parts.

(b) Table listing wrench sizes and assembly torques (or other equivalent procedures) for making up all joints and threaded assemblies.

(c) Instructions to permit overhaul by shipyard or other repair facility. These should include procedures for checking all critical dimensions subject to wear or change and the acceptable dimensional limits, surface finish condition, etc. Also, the appropriate procedure (that is, part replacement, correction at repair facility, or repair at manufacturer's facility) which should be followed to correct each case of damage or wear.

(d) Detailed disassembly and reassembly procedures. In addition to providing procedures for the complete disassembly and reassembly of the equipment, maintenance and troubleshooting sections shall contain, or refer to, only the limited disassembly and reassembly required to accomplish each particular operation. This is intended to reduce the possibility of unnecessary disassembly and unnecessary disturbance of adjustments when performing specific or limited maintenance or troubleshooting operations.

(e) Adjustment procedures.

☆U.S. GOVERNMENT PRINTING OFFICE: 1990 - 704-034/30886
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:39Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-G-24745(SH)	

3. DOCUMENT TITLE GOVERNORS, STEM DRIVEN PUMP, PRESSURE REGULATING, NAVAL SHIPBOARD, 600 POUNDS PER SQUARE INCH STEAM SERVICE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Point of Contact (TPOC): HOWARD BROWN PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial 703 602-9711 TPOC: SEA 56Y22	(2) AUTOVON 332-9711
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy (SEA 55Z3) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 Previous editions are obsolete. 198/290

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:39Z
Check the source to verify that this is the current version before use.